

Agentic Wealth Management

Centralized & Distributed Memory in Multi-Agent AI Systems



A multi-agent AI pipeline using LangGraph — mapping to Microsoft Azure enterprise architecture

Agent Memory Architecture



CENTRALIZED SHARED MEMORY

LangGraph Shared State · SQLite Client Database

Client Profile

Portfolio Holdings

Tax Regime

Transactions

Annual Income

Data Retrieval Agent

Fetches & structures
client portfolio data

Tax Analysis Agent

Applies IT Act 2025
& flags opportunities

Advisory Agent

Generates strategy
& recommendations

DISTRIBUTED (AGENT-PRIVATE) MEMORY

DB Schema Memory

- Schema definitions
- SQL query templates
- Field type mappings
- Data transformations

Tax Domain Memory

- IT Act 2025 rules
- Section 80C / 80CCD
- LTCG & STCG limits
- Capital gains tables

Market Domain Memory

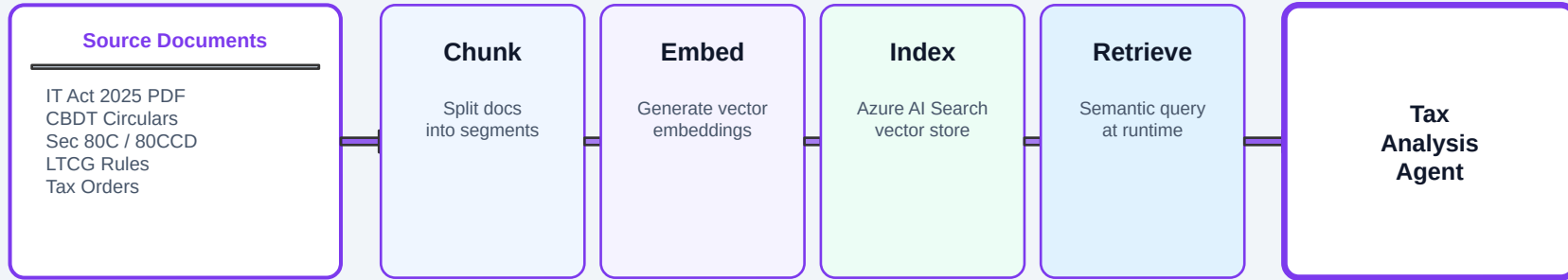
- Asset benchmarks
- RBI / SEBI guidelines
- Fund performance data
- Rebalancing models

Distributed Memory: RAG & Indexing Architecture

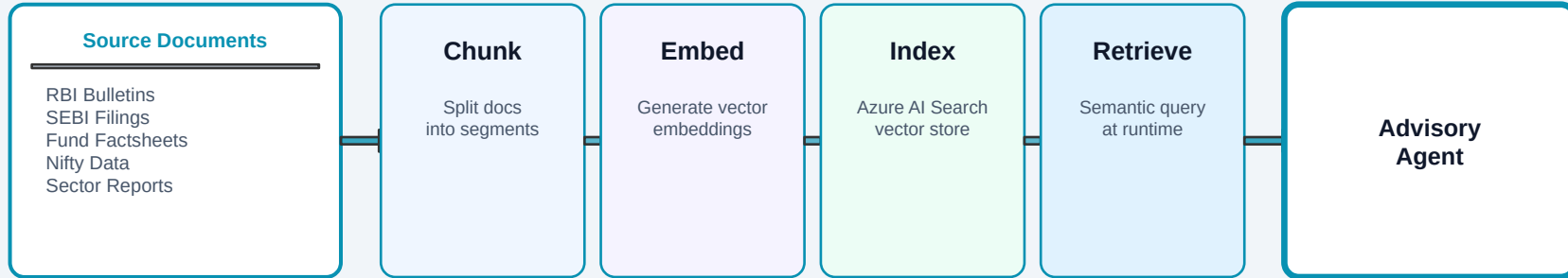


How each agent retrieves its private domain knowledge at query time — prototype path and Azure enterprise path

TAX AGENT — DOMAIN KNOWLEDGE (IT ACT 2025)



ADVISORY AGENT — DOMAIN KNOWLEDGE (MARKET & REGULATORY)



Microsoft Ecosystem Mapping



From open-source prototype to enterprise-grade Azure deployment — a clear transition path for each architectural component

COMPONENT	PROTOTYPE (DEMO)	MICROSOFT ENTERPRISE
Orchestration	LangGraph Python (agent wiring & shared state)	 Azure AI Foundry (pro-code agent orchestration & governance)
User Interface	Streamlit (developer dashboard)	 Microsoft Copilot Studio (low-code conversational front door)
LLM Serving	Ollama + Kimi K2.6 (local & OpenClaw gateway)	 Azure Model Catalog (GPT-4o, Phi-4, open-weight models via API)
Centralized Memory	SQLite + LangGraph shared state object	 Microsoft Fabric / Azure SQL + Foundry pipeline state
Distributed Memory	Domain rules embedded in agent prompts	 Azure AI Search (per-agent isolated vector indexes)
Response Cache	JSON file cache (Fast Mode toggle)	 Azure Cache for Redis (sub-10ms cached retrieval)
Security & Compliance	Local only (no auth layer)	 Microsoft Entra ID + Purview (RBAC, audit trail, data lineage)
Observability	Agent Timeline UI (Streamlit in-app view)	 Azure Monitor + App Insights (distributed tracing, dashboards)

Client Reporting Automation Agent



Generates personalized client reports at scale — what took a 3-day manual effort now takes 20 minutes across thousands of portfolios

INPUT

- Client portfolio data
- Benchmark & NAV feeds
- Reporting period trigger
- RM customization flags

Client Reporting Agent

OUTPUT

- Personalized PDF report
- RM review queue entry
- Auto-distribution to client
- Audit log entry

MEMORY ARCHITECTURE

CENTRALIZED SHARED MEMORY

- Client profile & communication prefs
- Portfolio holdings & weightings
- Transaction & SIP history
- Performance benchmarks (YTD / 1Y / 3Y)
- Tax summary & realized gains / losses

DISTRIBUTED MEMORY · RAG

- Report narrative templates (HNI / retail)
- Regulatory disclosure language
- Market commentary library (sector-wise)
- Fund performance & star rating data
- Benchmark index definitions (Nifty / Sensex)

Trade Reconciliation Agent



Matches executed trades against custodian records end-of-day — auto-resolves common breaks, surfaces only genuine exceptions to ops team

Historical context: Among the earliest AI use cases in capital markets — rule-based matching engines date to the 1990s. Modern LLM agents now handle unstructured exceptions and narrative break descriptions that rigid rules never could.

INPUT

- EOD trade blotter
- Custodian settlement records
- Exchange confirmations
- ISIN master reference

Trade Reconciliation Agent

OUTPUT

- Matched trade register
- Break report with severity
- Auto-resolved items log
- Exception queue for ops

MEMORY ARCHITECTURE

CENTRALIZED SHARED MEMORY

- Internal trade blotter (all executed orders)
- Custodian & depository records
- Client account positions
- Settlement status & funding state
- Historical break resolution log

DISTRIBUTED MEMORY · RAG

- Reconciliation rule playbooks
- SEBI settlement cycle guidelines (T+1)
- NSE / BSE / CDSL exchange norms
- Break resolution decision tree
- Common exception pattern library

AI Voice Support Assistant



Handles inbound client calls autonomously — portfolio queries, SIP changes, redemptions — escalates to human RM only on genuine exceptions

INPUT

- Inbound voice call (telephony)
- Speech-to-text transcript
- Client ID & auth token
- Intent classification

Voice Support Agent

OUTPUT

- Voice response (TTS)
- Action executed (SIP / redemption)
- Escalation to live RM
- CRM ticket created

MEMORY ARCHITECTURE

CENTRALIZED SHARED MEMORY

- Client account & KYC data
- Portfolio holdings & live NAV
- Open service tickets & history
- SIP schedules & redemption status
- Transaction & statement history

DISTRIBUTED MEMORY · RAG

- Product FAQ & knowledge base
- Fund / bond / FD catalog
- Compliance call scripts & disclosures
- Escalation decision tree
- SEBI investor grievance guidelines

Regulatory Compliance Agent



Scans all portfolios daily for SEBI / AMFI violations, insider-watch breaches, and concentration limit failures — zero manual compliance monitoring

INPUT

- Live portfolio positions
- Market data & NAV feed
- Regulatory update alerts
- Insider watch-list changes

Regulatory Compliance Agent

OUTPUT

- Daily compliance dashboard
- Violation alerts to officer
- Regulatory filing drafts
- Audit trail (Azure Purview)

MEMORY ARCHITECTURE

CENTRALIZED SHARED MEMORY

- All client portfolio positions
- AUM & NAV data (all funds)
- Client classification (HNI / retail / institutional)
- Historical trade data (3-year lookback)
- Risk category & mandate restriction flags

DISTRIBUTED MEMORY · RAG

- SEBI regulations & latest circulars
- AMFI guidelines & concentration limits
- Insider trading watch-list (NSE / SEBI)
- PMLA & KYC compliance rules
- Mandate-specific restriction tables

More Agentic Finance Use Cases



Each use case below follows the same memory pattern — centralized client data + distributed domain knowledge — no additional architecture needed

01

Intelligent Client Onboarding

Conversational KYC collection, PAN/CIBIL verification, and risk profiling. Auto-populates CRM. RM handoff only at final sign-off — reduces onboarding from days to minutes.

02

Real-Time Portfolio Q&A

"What is my LTCG exposure if I sell today?" — Agent fetches live holdings, applies IT Act rules from RAG memory, and answers in seconds. No advisor needed for routine queries.

03

Tax-Loss Harvesting Agent

Nightly scan of all portfolios for unrealized losses that can offset LTCG. Flags 80C/80CCD shortfalls before fiscal year-end. Generates harvesting plan with projected savings.

04

Portfolio Rebalancing Agent

Monitors allocation drift daily. When equity/debt/gold drifts beyond threshold, proposes a rebalancing trade set with full tax impact — sent for one-click RM approval.

05

Document Intelligence Agent

Continuously ingests fund factsheets, SEBI circulars, DRHP filings, and annual reports. Chunks, embeds, and indexes into per-fund isolated RAG vector stores — powering all other agents.